QUBO Formulation: Planar Robot Inverse Kinematics Candidate Selection

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Problem Statement. Given a planar robot with N discretized joint-angle candidates, the task is to select exactly one candidate that minimizes the combined inverse kinematics (IK) error and regularization cost. Let $N \in \mathbb{Z}^+$ denote the number of candidates, and let $c_i \in \mathbb{R}$ denote the cost associated with candidate i for $i \in \{0, \ldots, N-1\}$. The goal is to identify the candidate index i^* that yields the minimum total cost while satisfying the selection constraint.

Decision Variables. We introduce binary decision variables $x_i \in \{0, 1\}$ for each candidate $i \in \{0, \ldots, N-1\}$. The variable x_i equals 1 if candidate i is selected, and 0 otherwise. The requirement that exactly one candidate must be chosen is enforced by the equality constraint $\sum_{i=0}^{N-1} x_i = 1$.

QUBO Derivation. **Step 1 — Original Objective.** The problem requires minimizing the total cost:

$$\min_{x \in \{0,1\}^N} \sum_{i=0}^{N-1} c_i x_i. \tag{1}$$

Step 2 — Constraint Formulation. The selection constraint is written algebraically as:

$$\sum_{i=0}^{N-1} x_i = 1. \tag{2}$$

Step 3 — Penalty Term Expansion. We convert the equality constraint into a quadratic penalty using a positive weight $\lambda > 0$:

$$P(x) = \lambda \left(\sum_{i=0}^{N-1} x_i - 1 \right)^2. \tag{3}$$

Expanding the square yields:

$$P(x) = \lambda \left(\sum_{i=0}^{N-1} x_i^2 + \sum_{i \neq j} x_i x_j - 2 \sum_{i=0}^{N-1} x_i + 1 \right). \tag{4}$$

Applying the idempotent property $x_i^2 = x_i$ for binary variables simplifies the expression to:

$$P(x) = \lambda \left(\sum_{i=0}^{N-1} x_i + \sum_{i \neq j} x_i x_j - 2 \sum_{i=0}^{N-1} x_i + 1 \right) = \lambda \left(\sum_{i \neq j} x_i x_j - \sum_{i=0}^{N-1} x_i + 1 \right). \tag{5}$$

Rewriting the pairwise sum over unordered indices gives the general penalty contribution:

$$P(x) = \lambda \left(2 \sum_{0 \leq i < j \leq N-1} x_i x_j - \sum_{i=0}^{N-1} x_i + 1 \right). \tag{6}$$

Step 4 — Combined QUBO Hamiltonian. Adding the objective and penalty terms produces the single QUBO Hamiltonian $H(x)$:

$$H(x) = \sum_{i=0}^{N-1} c_i x_i + \lambda \left(2 \sum_{0 \leq i < j \leq N-1} x_i x_j - \sum_{i=0}^{N-1} x_i + 1 \right). \tag{7}$$

Grouping coefficients by variable products yields:

$$H(x) = \sum_{0 \leq i < j \leq N-1} (2\lambda) x_i x_j + \sum_{i=0}^{N-1} (c_i - \lambda) x_i + \lambda. \tag{8}$$

Step 5 — Q Matrix Entries and Offset. Reading off the coefficients from the general closed form gives:

$$Q_{i,j} = 2\lambda, \qquad 0 \leq i < j \leq N-1, \tag{9}$$

$$Q_{i,i} = c_i - \lambda, \qquad 0 \leq i \leq N-1, \tag{10}$$

$$c = \lambda. \tag{11}$$

Penalty Weight Justification. We require $\lambda > \sum_{i=0}^{N-1} |c_i|$ to ensure that any violation of the equality constraint incurs a penalty strictly greater than the maximum possible reduction in the objective function. Specifically, if a candidate assignment violates $\sum_i x_i = 1$, the squared term $\left(\sum_i x_i - 1\right)^2$ is at least 1 for integer variables, incurring a minimum penalty of λ. The maximum possible gain from the objective function by selecting an invalid configuration is bounded by $\sum_i |c_i|$. Thus, choosing $\lambda > \sum_i |c_i|$ guarantees that infeasible solutions always have higher energy than the optimal feasible solution.

Correctness Argument. Minimizing $H(x)$ over $\{0,1\}^N$ recovers the optimal candidate selection because the penalty term is zero if and only if $\sum_i x_i = 1$, and strictly positive otherwise. For any feasible assignment, $H(x)$ reduces exactly to the original objective $\sum_i c_i x_i$. For any infeasible assignment, the energy exceeds the minimum feasible energy by at least $\lambda - \sum_i |c_i| > 0$. Consequently, the global minimum of $H(x)$ must correspond to a feasible assignment that minimizes the original cost, yielding both the optimal candidate index and the minimum cost.

Illustrative Example. Consider an instance with $N = 3$ candidates and costs $c_0 = 2.5$, $c_1 = 1.8$, $c_2 = 3.2$. Let $\lambda = 10$. The objective is $2.5x_0 + 1.8x_1 + 3.2x_2$. The penalty term expands to $10(2x_0x_1 + 2x_0x_2 + 2x_1x_2 - x_0 - x_1 - x_2 + 1)$. Combining these, the concrete QUBO Hamiltonian is:

$$\begin{aligned} H(x) &= 20x_0x_1 + 20x_0x_2 + 20x_1x_2 + (2.5-10)x_0 + (1.8-10)x_1 + (3.2-10)x_2 + 10 \quad (12) \\ &= 20x_0x_1 + 20x_0x_2 + 20x_1x_2 - 7.5x_0 - 8.2x_1 - 6.8x_2 + 10. \quad (13) \end{aligned}$$

Evaluating the three feasible solutions ($x_0 + x_1 + x_2 = 1$):

$$\begin{aligned} x &= (1,0,0) \Rightarrow H = 2.5, \quad (14) \\ x &= (0,1,0) \Rightarrow H = 1.8, \quad (15) \\ x &= (0,0,1) \Rightarrow H = 3.2. \quad (16) \end{aligned}$$

The minimum energy is 1.8, achieved at $x^* = (0,1,0)$, corresponding to candidate 1 with minimum cost 1.8.